

Unidad IV: Cantidad de movimiento.

$$p = m v$$

$$F^{\text{ext}} = m a$$

$$F^{\text{ext}} = m \cdot \frac{dv}{dt}$$

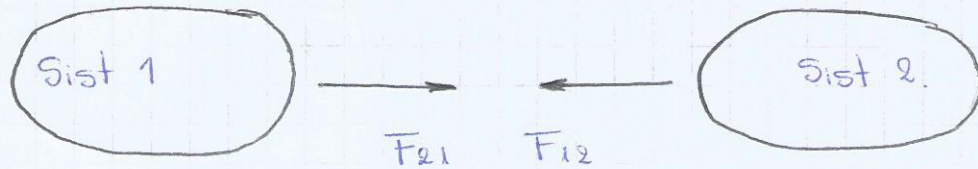
$$F^{\text{ext}} = \frac{d(m v)}{dt}$$

$$F^{\text{ext}} = \frac{dP}{dt}$$

$$\text{Si } F^{\text{ext}} = 0$$

entonces:

$$P = \text{cte.}$$



$$F_{12} = -F_{21}$$

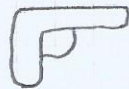
$$\frac{dP_2}{dt} = -\frac{dP_1}{dt}$$

$$\frac{d}{dt} (P_2 + P_1) = 0$$

Ejemplo:



Sistema 1



Sistema 2

$$P_i = P_A + P_B$$

$$P_i = 0 + 0$$

$$P_i = 0$$

$$P_f = P_A + P_B$$

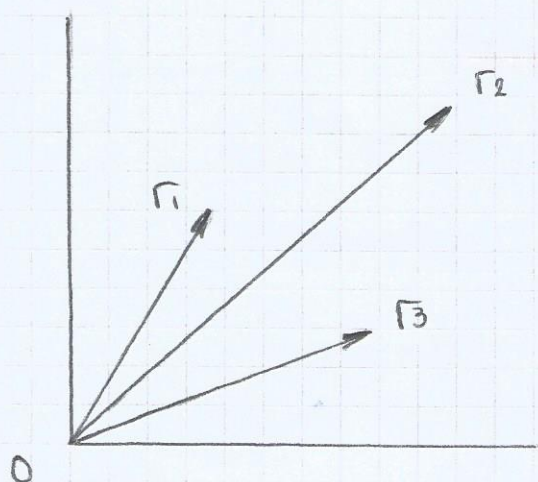
$$0 = P_A + P_B$$

$$P_A = -P_B$$

$$m_a v_a = -m_b v_b$$

$$v_a = \frac{-m_b v_b}{m_a}$$

Sistema de partículas:



$$P = \sum_{i=1}^N m_i \vec{v}_i$$

$$P = \sum_{i=1}^N \vec{p}_i$$

$$\frac{dP}{dt} = \sum_{i=1}^N \frac{d\vec{p}_i}{dt} = \sum_{i=1}^N m_i \vec{a}_i$$

$$\sum_{i=1}^N \vec{F}_i^{\text{ext}}$$

$$\sum_{i=1}^N \frac{d\vec{p}_i}{dt} = \vec{F}_i^{\text{ext}} + \sum_{j \neq i} \vec{f}_{ji}$$

$$\frac{dP}{dt} = \sum_{i=1}^N \vec{F}_i^{\text{ext}} + \underbrace{\sum_{i=1}^N \sum_{j \neq i} \vec{f}_{ji}}_{\vec{f}_{ij} + \vec{f}_{ji} = 0} = \sum_{i=1}^N \vec{F}_i^{\text{ext}}$$

$$P = M \cdot \vec{v}$$

$$P = \sum_{i=1}^N m_i \vec{v}_i$$

$$\text{con } M = \sum_{i=1}^N m_i$$

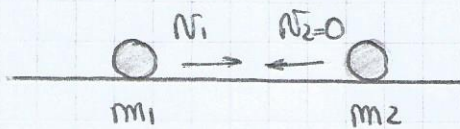
$$\vec{R}_{\text{cm}} = \frac{1}{M} \sum_{i=1}^N m_i \vec{r}_i$$

$$M \vec{R}_{\text{cm}} = \sum_{i=1}^N m_i \vec{r}_i$$

$$\begin{aligned}
 P &= M \cdot v_{cm} \\
 F_{ext} &= M \cdot a_{cm} \\
 P'_{cm} &= 0 \\
 P &= 0
 \end{aligned}
 \left. \vphantom{\begin{aligned} P &= M \cdot v_{cm} \\ F_{ext} &= M \cdot a_{cm} \\ P'_{cm} &= 0 \\ P &= 0 \end{aligned}} \right\} \text{ Sistema centro de masa}$$

$$\begin{aligned}
 P' &= \sum_{i=1}^N m_i v_i' = \sum_{i=1}^N m_i (v_{cm} - v_i) \\
 &= \sum_{i=1}^N m_i v_{cm} - \sum_{i=1}^N m_i v_i \\
 &= M v_{cm} - P = 0
 \end{aligned}$$

Sistema Laboratorio:



$$P_i = m_i v_i$$



$$P_f = (m_1 + m_2) v_f$$

$$P_i = P_f$$

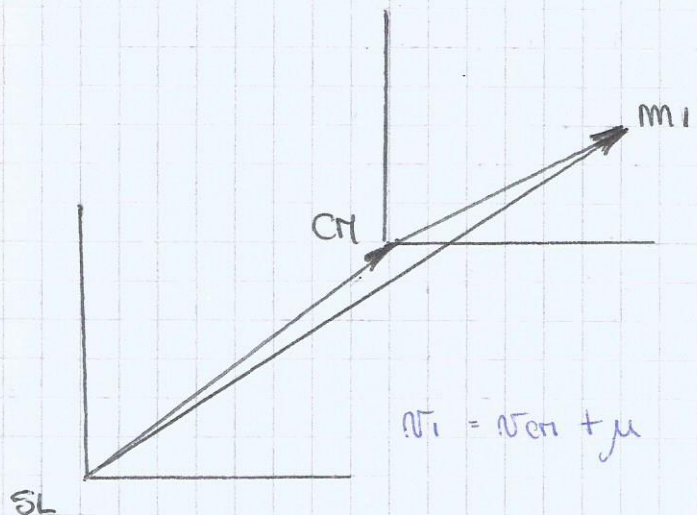
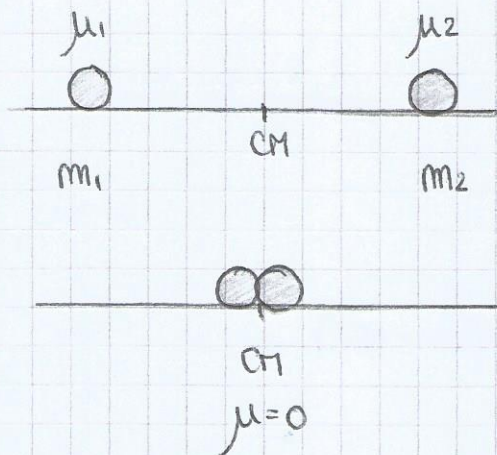
$$m_1 v_1 = (m_1 + m_2) v_f$$

$$v_f = \frac{m_1 v_1}{m_1 + m_2}$$

$$P = (m_1 + m_2) v_{cm} = m_1 v_1$$

$$v_{cm} = \frac{m_1 \cdot v_1}{m_1 + m_2}$$

Sistema centro de masa:

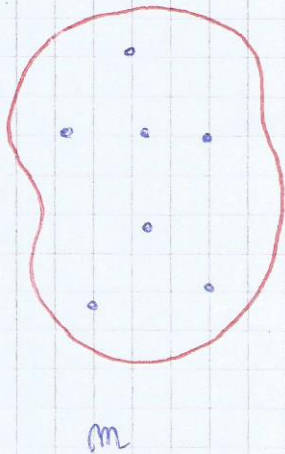


Antes: $\mu_1 = v_1 - v_{CM} = v_1 - \frac{m_1 v_1}{m_1 + m_2} = \frac{m_2 v_1}{m_1 + m_2}$

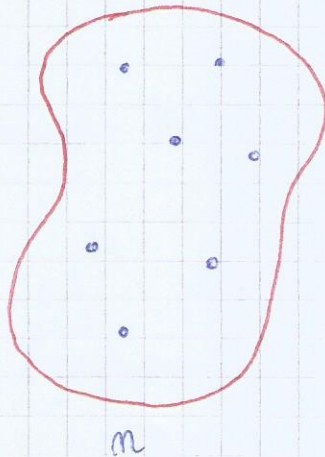
$\mu_2 = -v_{CM} = -\frac{m_1 v_1}{m_1 + m_2}$

Después: $\mu = 0$

Sist 1



Sist 2



N partículas

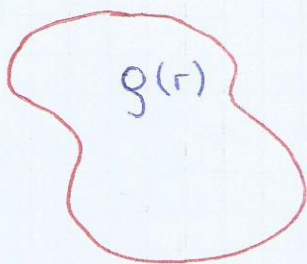
$N = m + n$

$M = M_1 + M_2$

$R_{CM} = R_{CM1} + R_{CM2}$

$M R_{CM} = \sum_{i=1}^N m_i r_i$

$M R_{CM} = \sum_{i=1}^N m_i r_i + \sum_{i=m+1}^{m+n} m_i r_i$



g (densidad de masa): cuanta masa hay por unidad de volumen

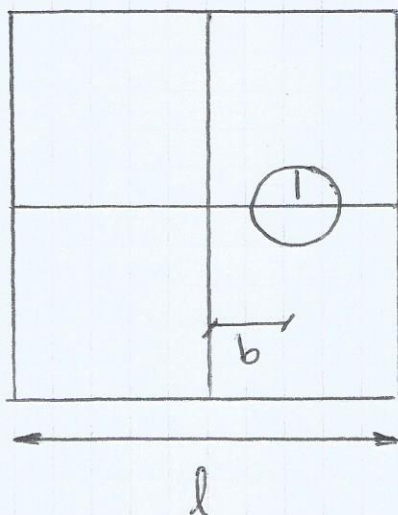
$$g = \frac{dm}{dv} = \frac{M}{V}$$

distribución
homogénea

$$M = \int_M dm = \int_V g(r) dv$$

$$R_{cm} = \frac{1}{M} \int_M r dm = \frac{1}{M} \int_V g(r) dv.$$

M



σ , densidad superficial

$$\sigma = \frac{dm}{dA} = \frac{M}{A} = M \int \sigma dA$$

uniforme

$$M R_{cm} = (M + m) R_1 - m R_2$$

$$M R_{cm} = -m b \hat{x}$$

$$R_{cm} = -\frac{m b}{M} \hat{x}$$

$$\sigma = \frac{m}{\pi a^2} \Rightarrow m = \pi a^2 \sigma$$

Masa variable:

$$F^{\text{ext}} = m \frac{dv}{dt} + v \frac{dm}{dt}$$

$$P_i = (M + m) v$$

$$P_f = (M + m - dm) (v + dv) + dm (v + dv - v_0) \quad (\text{respecto de T})$$

Con $(v + dv - v_0)$ velocidad dm respecto de Tierra

$$dp = P_f - P_i = (M + m) dv - dm v_0$$

$$F^{\text{ext}} = \frac{dP}{dt} = (M + m) \frac{dv}{dt} - v_0 \frac{dm}{dt} = 0$$

$$a = \frac{v_0}{M + m} \cdot \frac{dm}{dt} \quad (\text{empuje})$$

